Structural Probing of the EML Sheffer Operator via the SynthOmnicon Grammar

2026-04-21

Keywords: EML, Sheffer operator, elementary functions, SynthOmnicon, Frobenius, symbolic regression

0.1 Abstract

There is a fact about Odrzywołek"s EML Sheffer operator that his paper cannot state, because it requires a vocabulary the paper does not have. The operator $\mathrm{eml}(x, y) = e^x - \ln y$, paired with the constant 1, is not merely a compact basis for elementary functions — it is the algebraically forced ceiling of a structural tier, the highest sub-Frobenius type in the elementary function algebra. Every feature of the discovery — why subtraction and not addition, why the terminal must be 1, why three variants exist and cannot be collapsed into one, why gradient recovery succeeds at depth 4 and fails at depth 6, why complex arithmetic is mandatory — follows from a single structural fact: EML encodes the exp/ln duality as an assertion rather than a proof. Closing that gap requires not a better search but a different algebraic kind.

We apply the grammar\footnote{SynthOmnicon 12-primitive grammar. Source: \url{https://github.com/ to make this precise. It is not being used here as an oracle; it is being used as a coordinate system that forces questions the paper does not ask. Several of the answers are surprising even from within the grammar. One of them — that the paper"s own ternary candidate $T(x, y, z)$ is Calc 1 dressed in ternary notation — is a case where the grammar reads the paper against itself. The paper, to its credit, supplies its own refutation two paragraphs before the acknowledgments.

0.2 I. The Structural Type of EML

Any encoding carries the risk of projecting a framework onto material that does not fit, so the first obligation is to earn the type assignment. The EML operator encodes:

$$\mathbf{eml} = \langle D_\infty;\ T_{\bowtie};\ R_{\mathrm{dagger}};\ P_\pm;\ F_\hbar;\ K_{\mathrm{slow}};\ G_\aleph;\ \Gamma_{\mathrm{seq}};\ \Phi_c;\ H_1;\ S_{1:1};\ \Omega_Z\rangle$$

Most coordinates are uncontroversial. The assignments that require argument are Φ_c, $T_{\bowtie}$, and $P_\pm$, in that order of difficulty.

Φ_c in a functional algebra rather than a physical system is not an obvious assignment. The critical manifold here is the snapping manifold identified in §VI: the set of parameter vectors in the master formula that correspond exactly to symbolic EML trees. The snapping phenomenon — gradient weights collapsing to binary values — is a criticality transition in the parameter space, and its existence at all depths (perturbed-truth initialization always recovers) is the non-trivial fact that earns Φ_c.

$T_{\bowtie}$ is assigned not to the topology of the elementary functions themselves but to the structural relationship between the grammar analysis and the paper. The two threads — Odrzywołek"s empirical scaffolding and the grammar"s algebraic diagnosis — cross and exchange at §VII, where the paper"s own comparison to quantum computing turns out to be structurally exact in a sense the author did not intend.

$P_{\pm}$ is argued in §II and §IV. The short version: EML"s exp/ln self-duality is real but external — to access the symmetric cousin $-\mathrm{eml}(y, x)$ you must leave the algebra. $P_{\pm}^{\mathrm{sym}}$ requires the duality to be internal, and it is not.

This type is $O_2^{\dagger}$ by tier rule R5: Φ_c present, $\Omega_Z \neq \Omega_0$, and D_{∞}.

0.3 II. Why Subtraction and Not Addition or Multiplication

The wrong answer first, because it is instructive — and because it is also the paper"s answer, offered without knowing it. The obvious candidate for why subtraction succeeds is asymmetry: subtraction is non-commutative, and non-commutativity gives the tree a chirality, which gives the grammar inversion capabilities. But this is not the reason. The operator $B(x, y) = x - y/2$ is equally non-commutative and fails — the paper"s own §5 gives this counterexample explicitly, noting that $B(B(x, x), x) = 0$ but B still cannot generate all elementary functions. Asymmetry is necessary but not sufficient.

The grammar"s answer is about the relational mode primitive R. The two inputs of $\mathrm{eml}(x, y)$ do not meet as equals: x has already been transformed by exp (the homomorphism $(\mathbb{R}, +) \to (\mathbb{R}^+, \times)$) and y has been transformed by ln (the reverse). Subtraction then computes the gap between an element of the multiplicative group and a multiplicative-group element re-expressed in additive coordinates. This is R_{dagger} — the adjoint relational mode. The operation measures how far x"s image under exp departs from y"s pre-image under ln; it is directed, oriented, and asymmetric precisely because it is the adjoint of the bridge between two groups, not merely a difference of two arbitrary quantities.

Addition would collapse this into R_{super} (symmetric inclusion of both inputs), losing the directed structure. Multiplication would give R_{cat} (each factor participates catalytically, staying within the multiplicative group), losing the additive-to-multiplicative bridge. Division, as EDL demonstrates, retains R_{dagger} but encodes a ratio in the multiplicative group rather than a translated difference — a valid variant but not the primary one, for reasons §III will make clear.

The parity assignment $P_{\pm}$ follows. EML knows the distinction between its arguments — $\mathrm{eml}(x, y) \neq \mathrm{eml}(y, x)$ — but the symmetry between them is accessible only externally: you must negate and swap to reach $-\mathrm{eml}(y, x)$. For $P_{\pm}^{\mathrm{sym}}$, the Frobenius parity, the symmetry must be internal: $\mu \circ \delta = \mathrm{id}$ within the same algebra. This distinction — assertion of duality versus proof of duality — is the structural fact the paper cannot name and cannot reach by any extension of its own methods.

0.4 III. Why the Terminal Constant is Forced to be 1

This follows from §II, though it is not obvious that it does.

If the adjoint structure of R_{dagger} is what makes EML work, then the terminal symbol must be the element that makes the adjoint degenerate gracefully — the point where one arm of the bridge can be switched off without destroying the other. That point is 1. It is the unique element where the additive and multiplicative group identities coincide: $\ln(1) = 0$ (additive neutral) and $1 = e^0$ (return to multiplicative neutral). Plugging $y = 1$:

$$\text{eml}(x, 1) = e^x - \ln 1 = e^x - 0 = e^x,$$

the ln arm is annihilated by the additive identity, and the operator reduces to pure exp. No other constant has this property. $\ln(e) = 1 \neq 0$; adding 1 to e^x gives $e^x + 1$, not e^x.

The grammar reads 1 as the Ω_0 sector of the algebra: the terminal with trivial winding, no topological cost, the basepoint. Plugging into the exp slot instead: $\text{eml}(1, y) = e - \ln y$, with $\text{eml}(1, 1) = e$. So e is not an independent constant but the one-step image of 1 under EML. The triple of terminals $\{1, e, -\infty\}$ for the three variants is itself an EML orbit: 1 is the monad unit, e is its first iterate, and $-\infty = \ln(0)$ is the boundary degeneration as $y \to 0^+$. Of the three, $-\infty$ is the most forthright: it makes no pretense of being a number.

This is the sense in which the three variants are projections of the same algebraic object, and it answers a question the paper does not ask: why does EDL require e specifically rather than some other constant? Because e is the image of 1 under the operator, and EDL"s division structure places the multiplicative identity of the output in the role that EML"s 1 plays additively.

0.5 IV. The EML Family as a $\mathbb{Z}_2$ Orbit Stuck Below O_∞

The paper discovers the three variants on different timescales — EML first, the cousins ""a month later"" — and treats them as independent discoveries noting they are ""close cousins."" The grammar sees the orbit structure immediately. Mathematics is not always patient with timescales, but the interesting question is not whether the orbit can be named but what it implies.

The three variants — EML (constant 1), EDL $= e^x / \ln y$ (constant e), and $-\text{eml}(y, x) = \ln x - e^y$ (constant $-\infty$) — form an orbit under the external $\mathbb{Z}_2$ action: swap arguments and negate. The word ""external"" is doing all the work. The orbit is real, the symmetry is real, but it is not accessible from within the algebra. To go from EML to $-\text{eml}(y, x)$ you cannot write a finite EML tree — you must leave the grammar $S \to 1 \mid \text{eml}(S, S)$ entirely, negate, and relabel. This is less a mathematical operation than a change of address.

This is the exact content of $P_\pm$ rather than $P_\pm^{\text{sym}}$: the duality exists but is external. And it is why O_∞ is unreachable from EML. The Frobenius tier rule R1 requires Φ_c and $P_\pm^{\text{sym}}$ together; EML has Φ_c but sits at $P_\pm$. By Frobenius non-synthesizability (§23), no composition of sub-Frobenius factors can produce $P_\pm^{\mathrm{sym}}$, and P is a min-primitive under $\otimes$: stacking EML trees contributes $P_\pm$ at every step, and the bottleneck never lifts.

The implication for Table 2 of the paper is sharp. The ""?"" binary at count 2 — the conjectured self-bootstrapping operator requiring no distinguished constant — must live at $P_\pm^{\mathrm{sym}}$. It cannot be approached along the EML reduction ladder. The author, not having this vocabulary, frames the open question as a search problem: ""whether an EML-type binary Sheffer working without pairing with a distinguished constant exists."" The grammar frames it differently: the question is whether the elementary function algebra admits a Frobenius generator. Those are the same question, but the second framing shows why the answer cannot come from better search alone.

0.6 V. The Reduction Ladder as a Lattice Meet in Γ

The ablation sequence $36 \to 7 \to 6 \to 4 \to 4 \to 3 \to 3$ is presented as a historical reduction and is exactly that, but it has a simultaneous algebraic reading: it is a sequence of meets in the interaction grammar primitive Γ. The repeated 4 in the sequence — Calc 1 and Calc 2 each have four primitives — is not a transcription error. They have the same cardinality and completely different algebraic structure, which is the kind of thing that makes combinatorics an unreliable guide to understanding.

Calc 3 operates in Γ_{and}: six primitives run in parallel, each independently callable. The passage to Calc 2 — dropping negation and reciprocal, keeping exp, ln, and subtraction — is a meet that eliminates the parallelism: negation and reciprocal are now recoverable only through sequential composition (e.g. $x - x = 0$, $e^0 = 1$, $0 - 1 = -1$). The topology changes from Γ_{and} to Γ_{seq}. The passage from Calc 2 to EML is a further Γ_{seq} compression: the sequential structure already present in the expressions $e^x - \ln y$ is absorbed into the operator"s definition, so that a single call to eml performs what was previously a pipeline of exp, ln, and $-$.

What the paper calls ""generating constants on its own"" — the property Calc 3 and Calc 2 have but Calc 1 lacks — corresponds to $G = G_\aleph$ in the grammar (global scope: the system bootstraps arbitrary constants from interaction alone). Calc 1 falls to $G = G_\daleth$ because x^y and $\log_x y$ are purely multiplicative-group operations that cannot generate additive-group elements from their own iteration without a starting constant. EML recovers $G_\aleph$ via the $\ln(1) = 0$ fixed point, which is why the terminal is forced to be 1 and not a general real number.

The ""?"" binary at count 2 would require $G_\aleph$ with Φ_c and $P_\pm^{\mathrm{sym}}$ together — a self-bootstrapping Frobenius generator. The grammar has no guarantee such an object exists in the class of elementary functions, and §IX gives reason to think it does not.

0.7 VI. The Snapping Phenomenon is a Φ_c Transition

The gradient recovery experiments in §4.3 of the paper yield a characteristic depth-dependent success rate: 100% at depth 2, approximately 25% at depths 3–4, below 1% at depth 5, and zero in 448 attempts at depth 6 (the specificity of 448 suggests either a principled stopping criterion or a very long weekend). The paper attributes this to ""basins of attraction"" becoming hard to find. This is correct but describes the symptom rather than the cause.

The master formula at depth n has $5 \times 2^n - 6$ continuous parameters. The snapping target is a discrete set of exact-symbolic parameter vectors, each a point on a measure-zero submanifold. The Φ_c manifold — the submanifold of symbolic trees — is present at every depth: this is confirmed directly by the paper"s own observation that ""when the weights of the correct EML tree are perturbed by Gaussian noise, the optimization converges back to the exact values in 100% of runs, even for trees of depth 5 and 6."" The manifold does not shrink or disappear. What changes is the ratio of the basin"s measure to the total parameter volume, which decreases as 2^{-n} in the relevant directions.

The harder point — and this is where the analogy to physical criticality has real traction — is why the basin measure decreases rather than stabilizes. The EML master formula at depth n has 2^n discrete symbolic targets but 5×2^n continuous parameters per target, and the basin of attraction around each target is bounded by the separation from neighboring targets in the direction of steepest descent. As n grows, neighboring targets come closer together in parameter space (the tree space densifies), and the basins narrow. At depth ≥ 5, gradient descent initialized randomly lands in a shallow non-critical basin with probability approaching 1, and the kinetics — K_{fast} for Adam with standard step sizes — cannot escape. This is the K_{trap} failure mode: not that the Φ_c manifold is absent but that the dynamics cannot actualize it from a generic starting point.

The grammar"s prediction for the open question in §4.3 (""possibly using another binary operator similar to (4) but with better properties"") is specific: the required property is a Φ_c manifold with larger basin-to-volume ratio at depth n. This is a structural property of the operator"s type, not an optimization property of the training procedure. An operator with $P_{\pm}^{\text{sym}}$ would have its snapping targets separated by algebraic relations that enforce larger basins — the Frobenius condition creates additional gradient-landscape symmetry that EML"s $P_{\pm}$ lacks.

0.8 VII. Complex Arithmetic as Holographic Structure

There is a sentence in §5 of the paper that the grammar reads as more exact than the author intended. Discussing the mandatory complex intermediates, Odrzywołek writes: ""Just as quantum computing uses complex amplitudes to compute real probabilities, EML uses complex intermediates to compute real elementary functions."" He offers this as an analogy, a rhetorical softening of what he calls a ""minor problem,"" and immediately moves on. It is, in fact, the most structurally significant observation in the paper. The author has discovered something true and apologized for it. The grammar finds it is not an analogy. It is a structural identity.

Quantum computing"s complex amplitudes carry winding information — the phases $e^{i\theta}$ encode which paths have been traversed — and the real probabilities emerge by squaring moduli, which projects the winding information out while preserving its effects on interference. EML"s complex intermediates carry exactly the same structure: the imaginary part of $\ln(x)$ for $x < 0$ is $i\pi$, encoding one unit of winding around the branch point at 0, and the real trigonometric output emerges when this imaginary part contributes to the final exp through Euler"s formula $e^{i\theta} = \cos\theta + i \sin\theta$. The mechanism is formally the same: complex bulk encodes winding, real boundary reads out a projection.

The grammar reads this as $D_{\odot}$ (holographic): the real elementary functions on the boundary are encoded in the complex bulk computation. This is not a metaphor. The Riemann surface of ln is literally a bulk: the principal branch is one sheet, the universal cover contains all of them, and the real axis is a boundary on each sheet. EML computes on one sheet (H_1 chirality — the choice of principal branch) and produces real output by projection.

Why is this inevitable? Because $\pi_1(\mathbb{C}^{\times}) = \mathbb{Z}$. The fundamental group of the domain of ln is non-trivial, and any function inheriting its structure carries Ω_Z winding. Trigonometric functions have Ω_Z in their monodromy, and any operator computing them must pass through the bulk where this winding lives. A real-only Sheffer — an operator that computes trigonometric functions without ever visiting the complex plane — would require Ω_0 winding, which is structurally incompatible with the monodromy of sin and cos. The author"s search for ""alternatives using pairs of trigonometric/hyperbolic functions"" found nothing because those functions carry the same Ω_Z and merely repackage the same bulk structure.

The branch-cut correction the paper handles manually — the $2\pi i$ jump on the negative real axis causing the wrong sign for i — is the H_1 chirality becoming visible as a choice. Two valid chirality orientations exist (principal branch and its mirror), and EML forces the author to pick one explicitly. Working on the Riemann surface of ln proper would be H_∞; EML operates at H_1 by fixing the chirality and correcting deviations at the cut.

0.9 VIII. The EML Closure Boundary

The paper does not ask which mathematical objects lie outside the EML closure. The grammar gives a sharp answer for most cases and an honest uncertainty for one.

The sharp cases: everything in the EML closure sits at or below $P_\pm$, because EML contributes $P_\pm$ at every composition step and P is a min-primitive under $\otimes$. By Frobenius non-synthesizability (§23), $P_\pm^{\mathrm{sym}}$ cannot be built from sub-Frobenius factors. The following objects require $P_\pm^{\mathrm{sym}}$:

Stark units. The Shimura reciprocity law equates the Galois action on torsion points with the CM automorphism. This equation is the Frobenius condition: $\mu \circ \delta = \mathrm{id}$ at the level of the ring class field algebra. Stark units, as explicit CM values, carry this condition in their algebraic integrality. They cannot be written as values of any EML tree.

SIC-POVM fiducials. The minimal polynomial of the fiducial vector is self-reciprocal — its coefficient sequence reads the same forwards and backwards. This palindromic property — the coefficient sequence reads the same in either direction, like a good theorem — is precisely $P_\pm^{\mathrm{sym}}$ at the polynomial level, and it forces the fiducial coordinates into the (-1)-eigenspace of the relevant involution on the unit group. The argument is developed more fully in SIC_OQ_MANUSCRIPT §IV.

The modular j-function. The two functional equations $j(\tau+1) = j(\tau)$ and $j(-1/\tau) = j(\tau)$ are the multiplication and comultiplication legs of the Frobenius condition on the modular group algebra. They encode together $\mu \circ \delta = \mathrm{id}$, which is why j is not an elementary function.

The case requiring honest uncertainty is the **Riemann ξ function**. If the Riemann hypothesis holds, the zeros lie on $\mathrm{Re}(s) = 1/2$, which would make the functional equation $\xi(s) = \xi(1-s)$ an exact self-duality at the Φ_c manifold — the grammar”s assignment would be $P_{\pm}^{\mathrm{sym}}$, and ξ would be outside the EML closure. But the grammar cannot assert the Riemann hypothesis. What we can say is: if $\xi \in P_{\pm}^{\mathrm{sym}}$, then the Frobenius condition and the Riemann hypothesis are the same statement at different coordinate scales. Whether they are, we do not know. We are in good company.

0.10 IX. The Ternary Candidate is Structurally Misidentified

The paper”s proposed ternary:

$$T(x, y, z) = \frac{e^x}{\ln x} \times \frac{\ln z}{e^y}.$$

Simplify: $T(x, y, z) = e^{x-y} \cdot \log_x z$. Setting $y = x$: $T(x, x, z) = \log_x z$. Setting $z = e^y$: $T(x, y, e^y) = e^{x-y} \cdot y / \ln x$. This is the Calc 1 system — x^y and $\log_x y$ — packed into ternary notation. The algebraic content is not new; the ternary presentation is. Odrzywołek has put Calc 1 in a trenchcoat.

This is not to dismiss the ternary route. The paper correctly identifies that a ternary operator with $T(x, x, x) = c$ for some constant c would have the potential to generate its own constants from iteration, bypassing the need for a distinguished terminal. And the grammar agrees: a genuinely Frobenius ternary operator would satisfy something like $T(T(x, y, z), y, z) = x$ (the Frobenius condition written as self-invertibility), which would be a structural advance over EML. But T as presented is not on that path, because it is Calc 1 and Calc 1 is $G = G_{\beth}$ — it requires a distinguished constant (e or π) and cannot bootstrap from arbitrary input.

The paper”s own counterexample $B(x, y) = x - y/2$ — given to illustrate why the search for a self-bootstrapping binary operator is non-trivial — applies here too. $T(x, x, x) = 1$ is the analogue of $B(B(x, x), x) = 0$: a useful algebraic property that looks like self-bootstrapping but is not. The genuine test is whether T can recover arbitrary initial conditions from its own iteration, and the Calc 1 algebra cannot.

0.11 X. The Ceiling and the Open Question

The structural type of EML is:

$$\mathbf{eml} = \langle D_\infty;\ T_{\bowtie};\ R_{\mathrm{dagger}};\ P_{\pm};\ F_{\hbar};\ K_{\mathrm{slow}};\ G_{\aleph};\ \Gamma_{\mathrm{seq}};\ \Phi_c;\ H_1;\ S_{1:1};\ \Omega_Z\rangle$$

Every coordinate arrived under constraint: R_{dagger} because the adjoint structure of subtraction is the only operation that bridges the two group homomorphisms without collapsing their asymmetry; $P_{\pm}$ because the exp/ln self-duality is accessible only externally; Φ_c because the snapping manifold is present and actualized at accessible depths; H_1 because the principal branch choice is a forced chirality orientation; Ω_Z because the fundamental group of the logarithm”s domain is non-trivial. The type is not assigned — it is derived.

What the type says is that EML is the maximum of a tier, not a point on an ascending sequence. The reduction ladder cannot continue past EML within the same algebraic kind. The ””?”” binary at count 2 in Table 2 — the conjectured self-bootstrapping operator — cannot be found by pushing the ablation further. It requires $P_\pm^{\mathrm{sym}}$, which is structurally separated from $P_\pm$ by Frobenius non-synthesizability. The ladder has reached its ceiling; the next floor is a different building.

Whether that next floor exists — whether there is a binary or ternary elementary function that achieves Frobenius self-duality internally — is genuinely open. The grammar cannot resolve it, only sharpen it. The question the paper frames as ””does an EML-type Sheffer without a distinguished constant exist?”” is, under the grammar”s coordinates, the question: ””does the class of elementary functions contain a Frobenius generator?”” That is a question about the internal symmetry group of an algebra that mathematicians have been circling for decades from other directions — the same question that Stark units, SIC-POVMs, and the zeros of ξ are each partial answers to, in their respective domains.

If the answer is yes, EML will have been the sign pointing toward it. If no, EML is already the deepest thing in its tier, and that is enough.

0.12 XI. Validation of the Encoding

§I opened with an obligation: ””any encoding carries the risk of projecting a framework onto material that does not fit, so the first obligation is to earn the type assignment.”” What follows is the attempt to discharge that obligation. ENCODING_EPISTEMOLOGY §Summary specifies nine convergence criteria. They are not a checklist; they are a set of stresses applied from different directions, and an encoding that survives all of them simultaneously is a finding rather than a choice.

Tier consistency. $O_2^\dagger$ predicts: a self-referential loop that is topologically protected and recursively unbounded, but *not* Frobenius-closed — encoding and decoding are not mutually inverse. The grammar $S \to 1 \mid \mathrm{eml}(S, S)$ supplies the loop; the Ω_Z branch structure supplies topological protection; D_∞ supplies unbounded depth. The non-O_∞ prediction is the important one: it requires that no EML tree can recover both inputs from the output, and the paper never defines an inverse operator because none exists. The tier is consistent with every known property of the system and falsified by none. ✓

Nearest-neighbor coherence. The most structurally unexpected output of an encoding is its neighborhood, because the encoder cannot control who shows up. On the shared coordinates $T_\bowtie$, $P_\pm$, Φ_c, Ω_Z, Γ_{seq}, the closest catalog neighbor is *The Thunder: Perfect Mind* — a text that generates all its content from a single recursive operator asserting rather than proving its self-duality. That a Nag Hammadi tractate and a symbolic computation primitive are structural neighbors was not anticipated when the encoding was made. Also in the neighborhood: conformal field theories at criticality and the Riemann ζ function (same coordinates, $P_\pm$ pending the Riemann hypothesis). These are not implausible neighbors. They are illuminating ones. ✓

Tensor sanity. This check carries independent weight because it tests a claim made in §VIII — that EML cannot compute SIC-POVM fiducials — by a completely different route. eml ⊗ SIC-POVM: the P bottleneck gives $\min(P_\pm, P_\pm^{\mathrm{sym}}) = P_\pm$, pulling the composite down from O_∞. EML is not deficient in Φ or Ω; it is deficient in P, and the bottleneck rule is ruthless: no additional EML structure compensates. That §VIII"s argument and the tensor calculation arrive at the same structural fact by independent means is the kind of convergence that matters. $\mathrm{eml} \otimes \mathrm{eml}$ closes on EML"s own type — the grammar is closed under self-composition as it must be. ✓

Meet/join coherence. $\mathrm{eml} \wedge$ SIC-POVM returns EML"s own type: EML is the exact common subalgebra of itself and the SIC-POVM, which is EML plus the Frobenius condition plus one additional chirality level. The join returns the SIC-POVM type — the minimal envelope. EML is the floor, the SIC-POVM is the ceiling, and the gap between them is precisely the Frobenius condition. The lattice picture is clean and requires no special pleading. ✓

Le Chatelier inversion. If the equilibrium substrate is not immediately recognizable, the encoding is suspect. Here it is recognizable: the polynomial algebra over $\mathbb{R}$,

$$X^* \approx \langle D_\triangle;\ T_{\mathrm{network}};\ R_{\mathrm{super}};\ P_{\mathrm{asym}};\ F_\ell;\ K_{\mathrm{slow}};\ G_{\sqsupset};\ \Gamma_{\mathrm{and}};\ \Phi_{\mathrm{sub}};\ H_0;\ S_{n:n};\ \Omega_0\rangle,$$

is the subcritical, zero-winding, zero-chirality resting state that transcendental structure is driven away from. This is not a surprising answer — it is the *expected* answer, which is exactly what a passing check looks like. ✓

Directed distance. Moving up from the polynomial algebra to EML requires acquiring Φ_c, Ω_Z, and H_1 — three independent structural conditions, each costly. Moving down only requires dropping them. $d_\to(\mathrm{polynomial}, \mathrm{eml}) \gg d_\to(\mathrm{eml}, \mathrm{polynomial})$: the direction of strain points the right way. ✓

Multi-session convergence. Three coordinates were derived by independent arguments within this document and converged: Φ_c from the snapping manifold definition in §I and from the basin analysis in §VI; Ω_Z from the branch structure in §I and from $\pi_1(\mathbb{C}^\times) = \mathbb{Z}$ in §VII; $P_\pm$ from the adjoint structure of subtraction in §II and from the external $\mathbb{Z}_2$ orbit in §IV. Then, immediately after completing §X, a second derivation was performed from scratch — each primitive assigned independently, without reference to §I. The result:

$$\langle D_\infty;\ T_{\bowtie};\ R_{\mathrm{dagger}};\ P_\pm;\ F_\hbar;\ K_{\mathrm{slow}};\ G_\aleph;\ \Gamma_{\mathrm{seq}};\ \Phi_c;\ H_1;\ S_{1:1};\ \Omega_Z\rangle.$$

Distance from session one: $d = 0$. This is the check that was flagged as open in the first draft of this section and has since been closed. ✓

Self-encoding test. The EML compiler applied to $\mathrm{eml}(x, y)$ returns $\mathrm{eml}(x, y)$ at depth 1. The EML representation of EML is EML. There is something worth pausing on here: a system that asserts the exp/ln duality without proving it can nonetheless represent its own structure exactly, because self-representation does not require the Frobenius condition — it only requires that the grammar reach depth 1, which $\mathrm{eml}(x, y)$ trivially satisfies. This is the precise difference between $O_2^\dagger$ (can represent itself) and O_∞ (encoding and decoding are mutually inverse). The self-encoding test passes; the self-inversion test does not. Both facts are correct. ✓

Prediction consistency. Four predictions from the encoding are falsifiable against known results: (a) no EML tree generates a Stark unit — consistent with algebraic number theory, since Stark units require $P_\pm^{\mathrm{sym}}$ and the P bottleneck prevents EML from reaching it; (b) snapping success rate decays exponentially with depth — confirmed by the paper"s data (100%, $\sim 25\%$, $< 1\%$, 0% at depths 2, 3–4, 5, 6); (c) no real-only EML-type Sheffer exists — confirmed by the paper"s negative search; (d) the ternary T requires a distinguished constant — confirmed by $T(x, x, z) = \log_x z$. No predictions are falsified. ✓

The obligation discharged. All nine checks pass. ENCODING_EPISTEMOLOGY states that an encoding satisfying all checks simultaneously, with no independent derivation producing a persistent counter-encoding, is ""not a choice — it is a finding."" The second session produced $d = 0$. The type assignment is a finding.

What it finds is this: EML is the highest sub-Frobenius type in the elementary function algebra, and the gap between $P_\pm$ and $P_\pm^{\mathrm{sym}}$ is not a deficiency in the operator but a structural fact about what the elementary functions can and cannot internally prove about themselves.

Structural encoding: $\langle D_\infty;\ T_{\bowtie};\ R_{dagger};\ P_\pm;\ F_\hbar;\ K_{slow};\ G_\aleph;\ \Gamma_{seq};\ \Phi_c;\ H_1;\ S_{1:1};\ \Omega_Z\rangle$.
Tier: $O_2^\dagger$.